RSHS

MATH O MAGIC

5·3

by

Greg Hurn

EDUCATIONAL DIRECTIONS PUBLICATIONS

INTRODUCTION

MATH O MAGIC 5·3 is part of a series of 12 books

MATHOMAGIC 1·2	MATHOMAGIC 4·3
MATHOMAGIC 1·3	MATHOMAGIC 4·4
MATHOMAGIC 2·3	MATHOMAGIC 5·3
MATHOMAGIC 2·4	MATHOMAGIC 5·4
MATHOMAGIC 3·3	MATHOMAGIC 6·3
MATHOMAGIC 3·4	MATHOMAGIC 6·4

These books are designed for students in years eight to ten.

1·2 and 1·3 are designed to revise the skills from earlier years and to be a platform from which high school skills are developed.

2·3, 3·3, 4·3, 5·3 and 6·3 develop skills in the SPACE, MEASUREMENT, and INFERENCE strands.

2·4, 3·4, 4·4, 5·4 and 6·4 address skills in the NUMBER and FUNCTION strands.

Stages 2, 3, 4, 5 and 6 indicate the level of difficulty, with each following in a sequential manner.

PREFACE

Often students find the necessary skill practice in Mathematics to be very boring. This series of books has been designed as a motivational tool to help the students to feel more enthusiastic about their work.

These worksheets are designed to be used in various ways –

1. MOTIVATION – Students are keen to solve the puzzle and keep asking for more.

2. INCENTIVE – As a reward for students who finish their work early.

3. SELF CHECKING
   – For students to be able to check their own work as they go along.

4. EXTRA PRACTICE
   – A fun way of doing extra practice on skills development.

5. HOMEWORK – an interesting way to get students to do more consolidation of mathematical skills in a way they enjoy.

6. REVISION – An exciting way to stimulate revision.

However you use these sheets, you will always be asked for more. The students enjoy working out the puzzles and riddles and get a great sense of satisfaction from getting them right.

Copies of this book may be obtained from:

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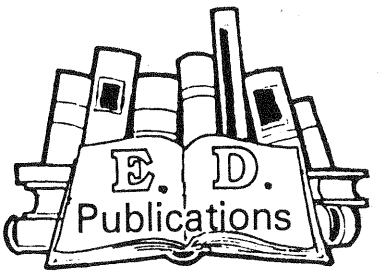
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Publishing editor – Greg Hurn

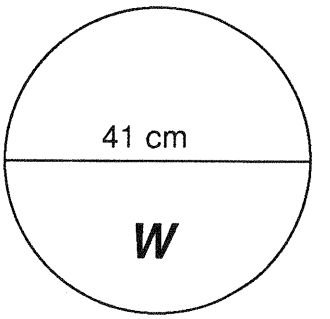
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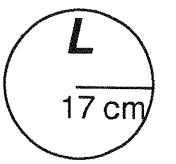


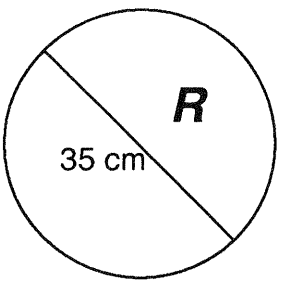


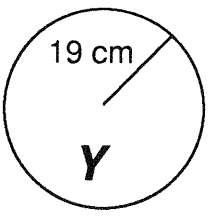
WHY ARE UNDERTAKERS LIKE TRUE FRIENDS?

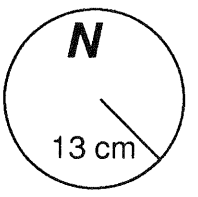
Find the circumference of each of the circles below.
(Use π ≈ 3·14159256 but round your answers to one decimal place.)

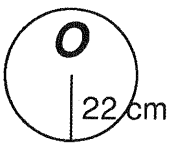


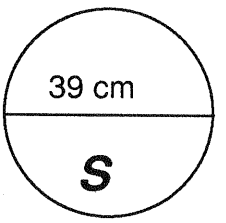


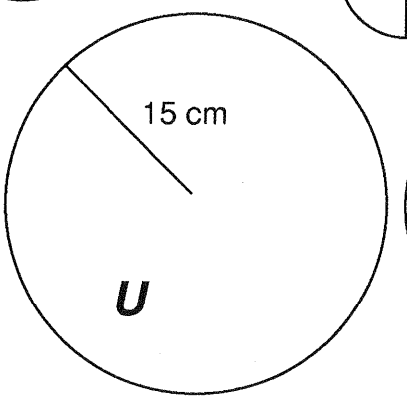


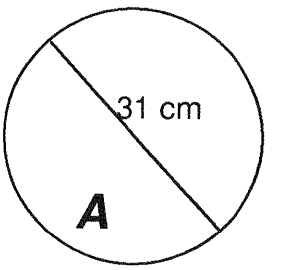


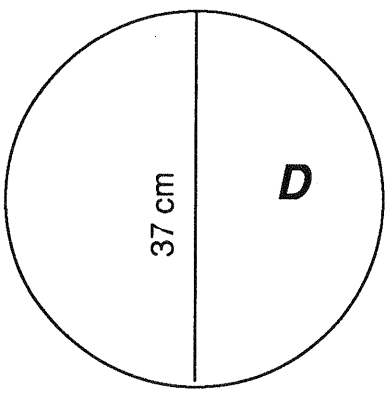


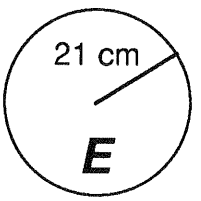


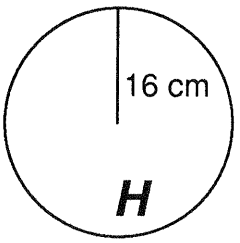


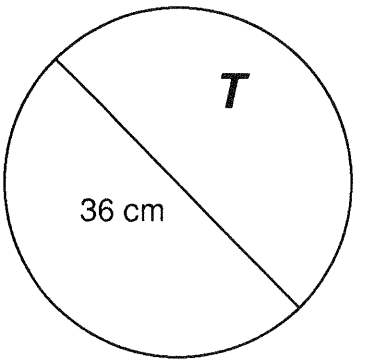










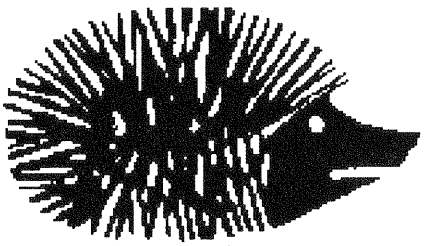


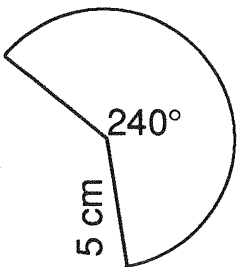
113·1	100·5	131·9	119·4	97·4	110	131·9	113·1	100·5	131·9	106·8	97·4	122·5	113·1	138·2	81·7	131·9	122·5
			113·1	138·2	106·8	131·9	113·1	119·4	138·2	94·2	116·2	138·2	128·8	81·7			

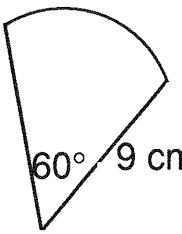
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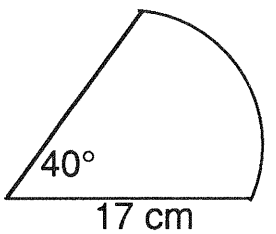
Find the AREA of each of the following SECTORS. Cross out the answers in the squares below. The remaining letters will tell you:
Use π ≈ 3·14159265
Round your answers to 1 d.p.)

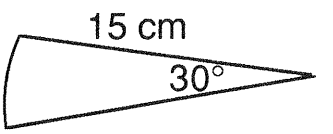
HOW DO YOU PICK UP A GUINEA PIG THAT'S BEEN CROSSED WITH A PORCUPINE?

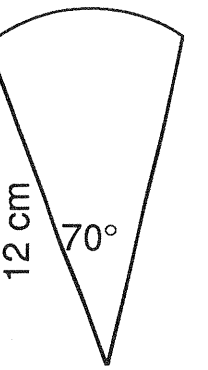


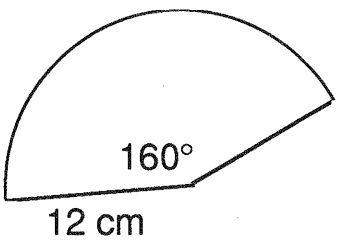


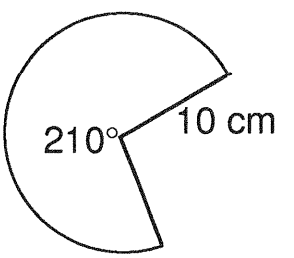


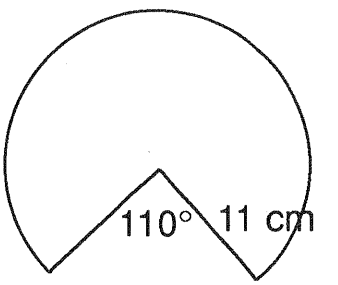


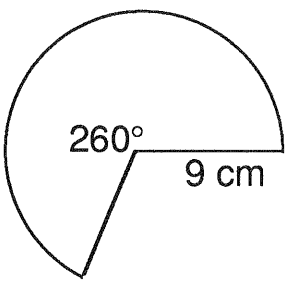


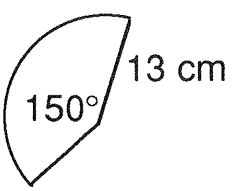


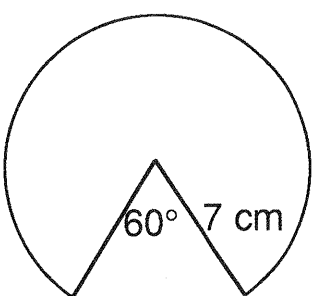










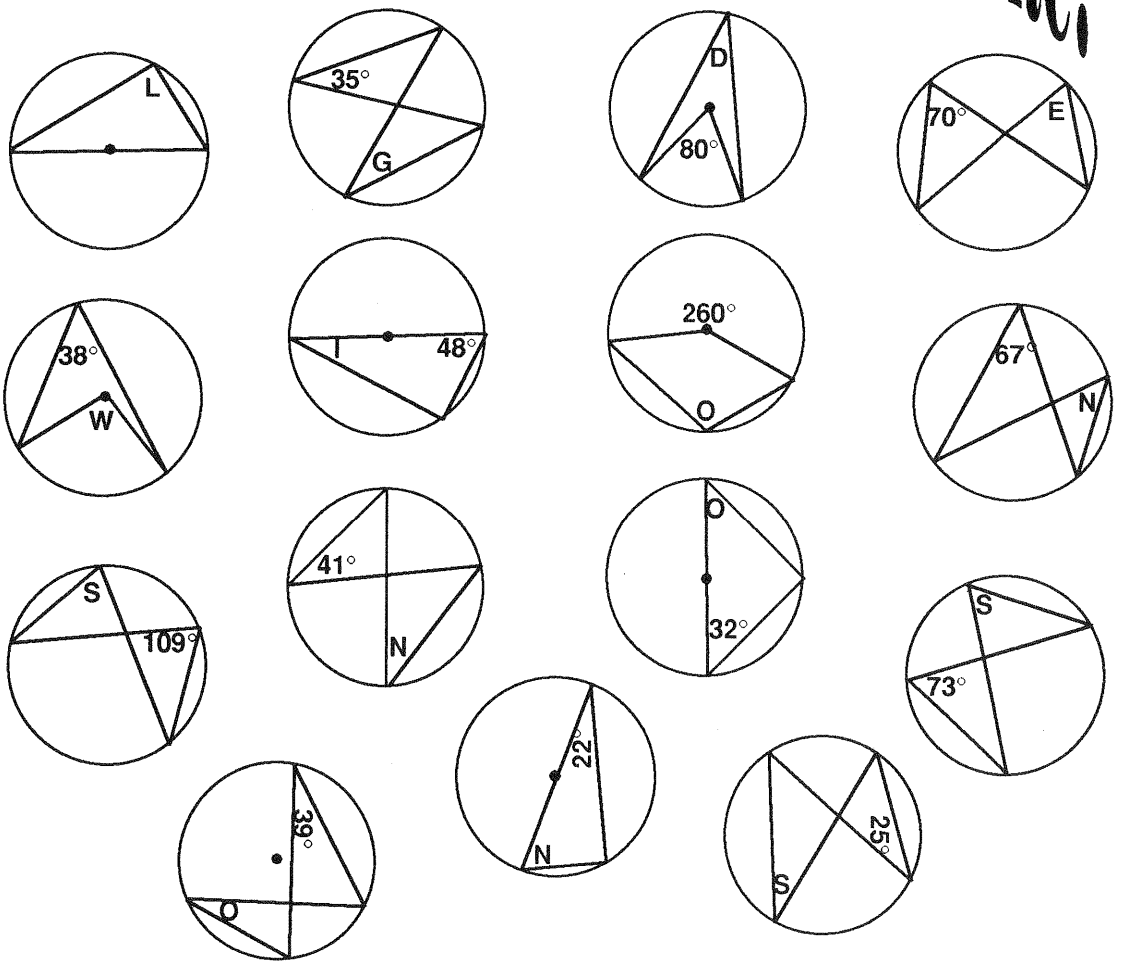


5	2·4	3	1	4	2·4	1	8	8	2	2	6	4	4·2	5	8·	9										
P	R	A	W	I	N	E	D	T	S	O	H	A	L	I	E	X	T	A	R							
1	0	0·9	6	1	2	8·3	9·7	1	8	3·3	7	8·	3													
T	O	M	O	T	Q	U	I	N	R	E	S	T	A	O	M	E	C									
2	2	1·2	1	8	3·8	6	7·4	2	0	1·1	9	1·	8													
B	E	E	T	M	W	O	N	A	U	T	U	A	T	O	I	O	N									

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What did the criminal say when he was saved from the hangman at the last minute?

Calculate the size of the indicated angle in each of the following circles. Use the answers to decode the puzzle.

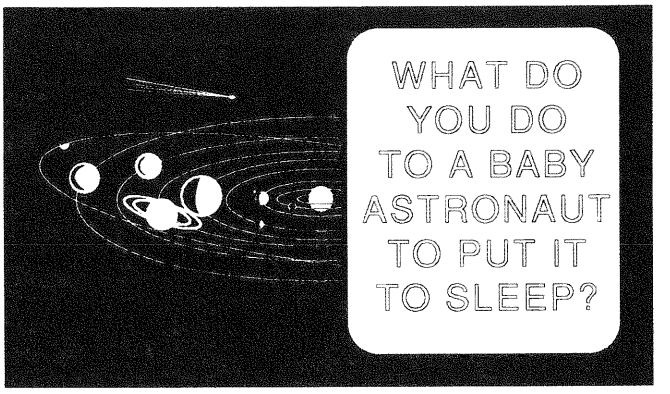


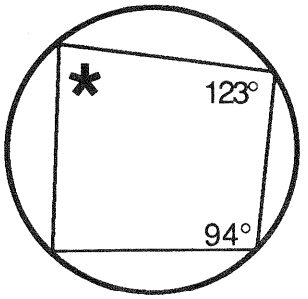
76°	70°	90°	90°	67°	39°	68°	39°	130°	25°	70°	42°	73°

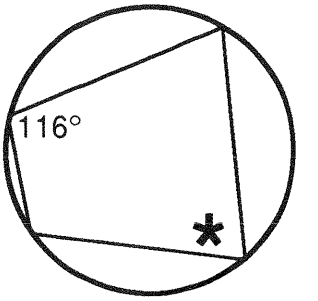
35°	130°	58°	40°	41°	58°	58°	109°	70°

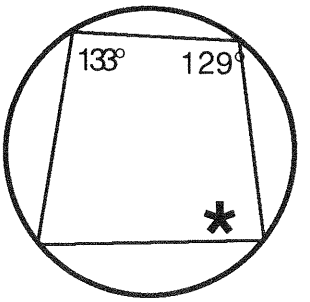
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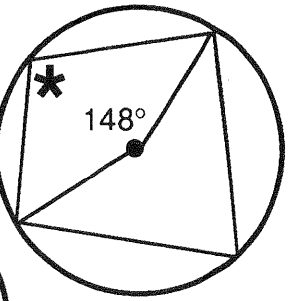
Find the size of the indicated angles. Cross out the letters attached to the answers to the problems below. The letters will tell you:

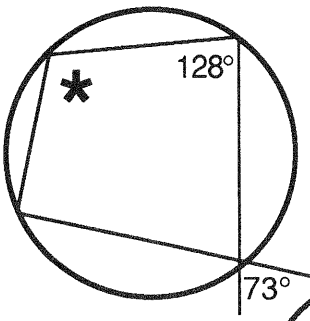


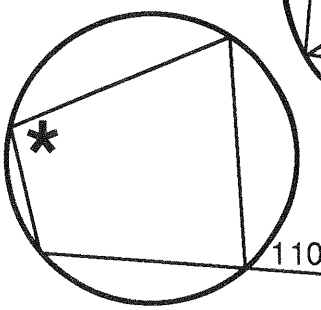


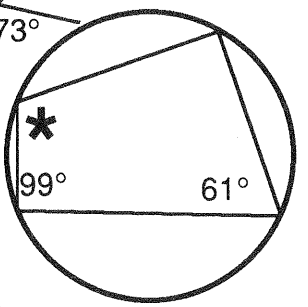


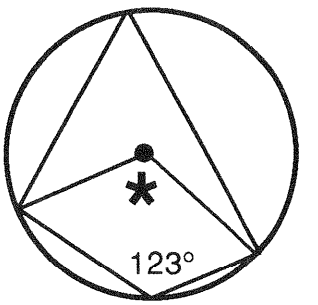


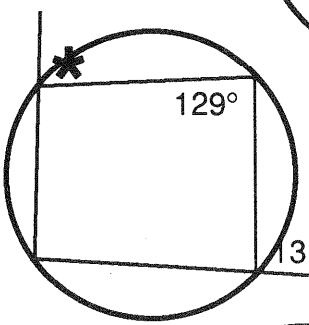


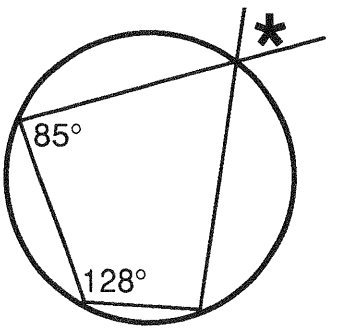


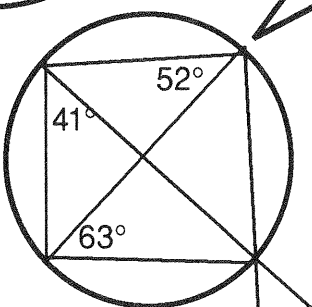


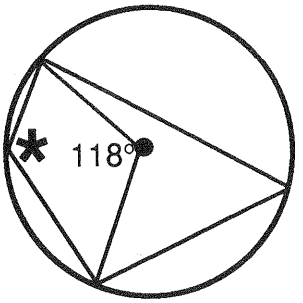




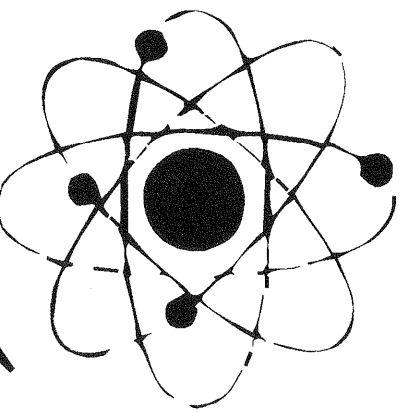








106°	A
47°	C
75°	Y
24°	R
96°	O
52°	A
86°	K
91°	U
119°	S
17°	R
62°	O
110°	T
131°	C
121°	N
97°	K
107°	M
127°	E
49°	O
64°	B
114°	A
84°	T



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Calculate the length of the arc in each of the following circles. Find the answers in the squares below. (Use π = 3·14159265 but round your answers to one decimal place.) This will tell you:



V
6 cm
175°

U
220°
7 cm

S
3 cm
260°

T
250°
10 cm

I
6 cm
235°

B
265°
9 cm

C
245°
7 cm

N
8 cm
255°

W
215°
9 cm

A
140°
15 cm

F
110°
13 cm

H
185°
6 cm

O
137°
16 cm

E
290°
8 cm

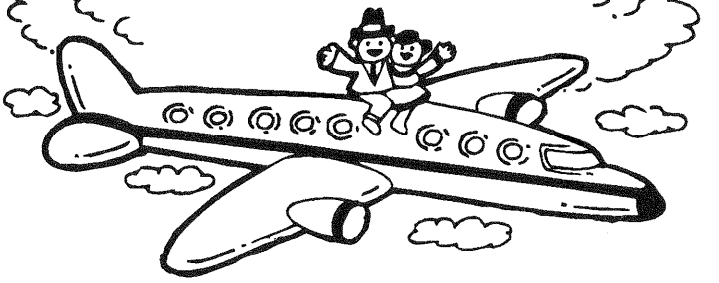
33·8	24·6	43·6	19·4

43·6	19·4	40·5	41·6	40·5	13·6	43·6	18·3	24·6	29·9	24·6	38·3	26·9	13·6

38·3	25	43·6	19·4	40·5	13·6	40·5	36·7	13·6	38·3	35·6

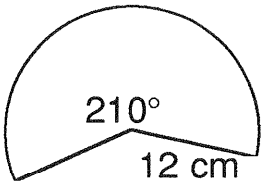
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WHAT WAS ON THE SIGN AT THE TRAVEL AGENCY?

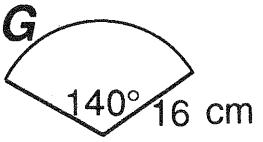


Find the perimeter of each of the following sectors.
(Use $\pi \approx 3{\cdot}14159265$ but round your answers to one decimnal place.)

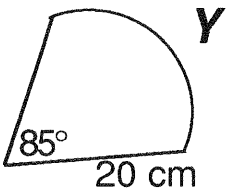
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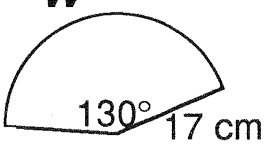
G



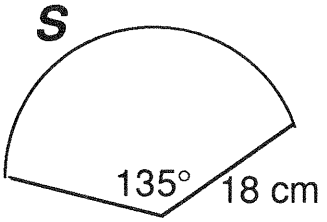
Y



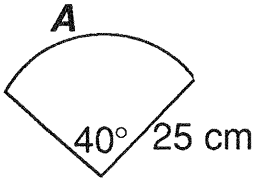
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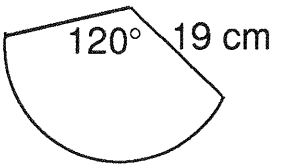
S



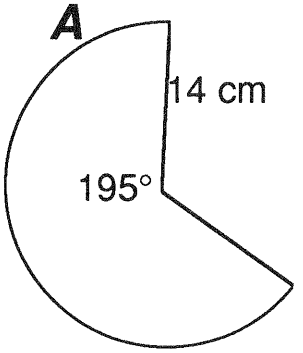
A



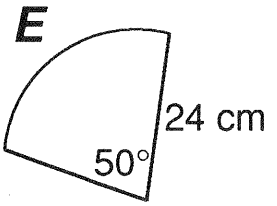
L



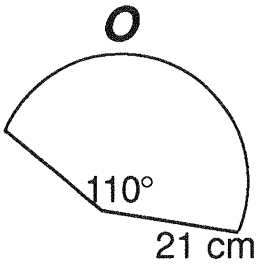
A



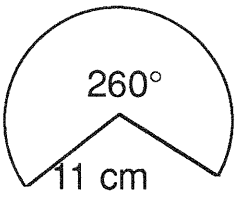
E



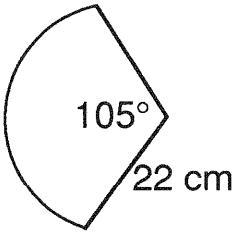
O



A



E



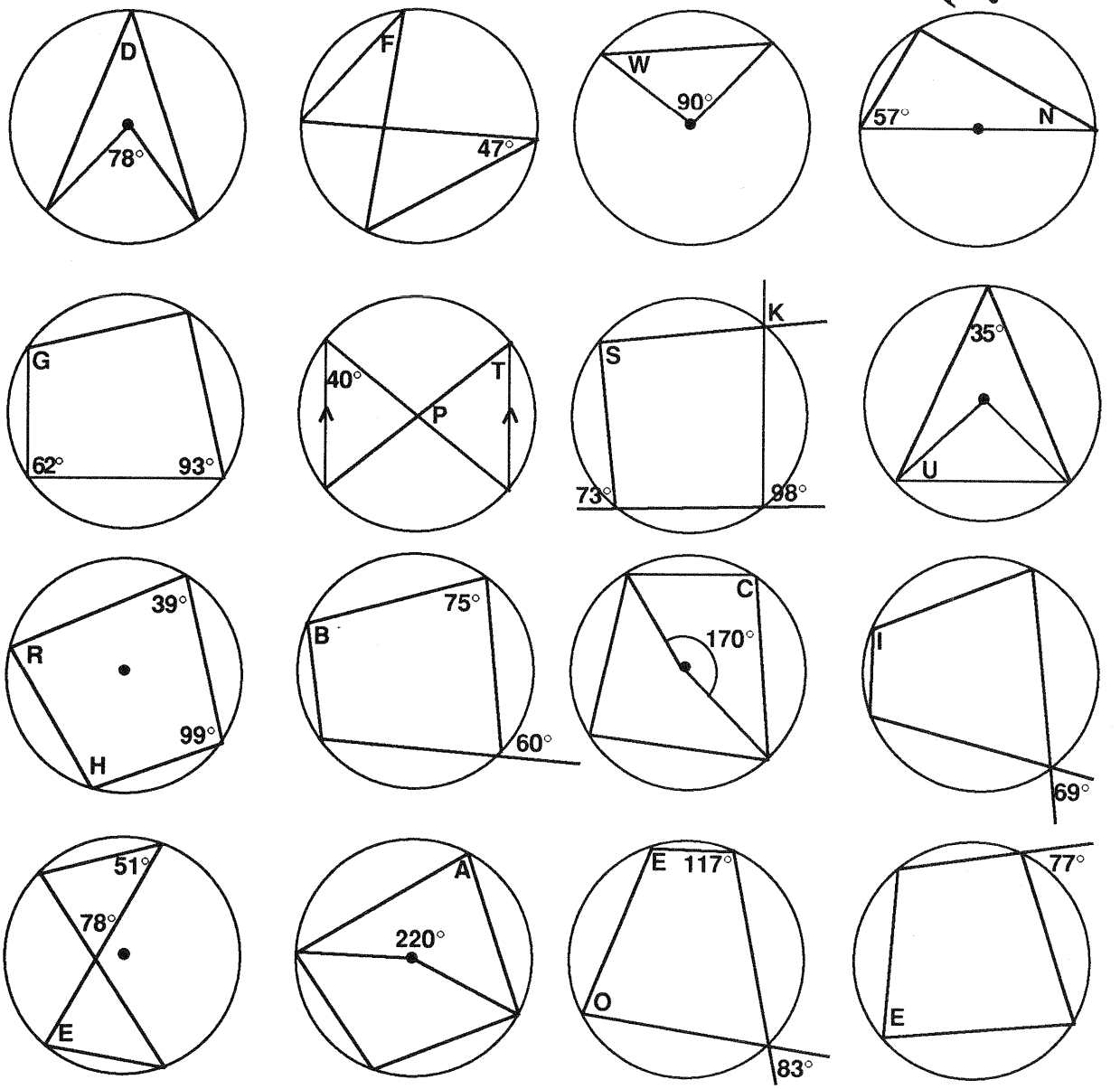
○ 68 ○ 77·8 ○ 68·9 ○ 71·9 ○ 78·4 ○ 84·3

○ 71·1 ○ 82·3 ○ 67·5 ○ 72·6 ○ 75·6 ○ 69·7

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Why Did Snoopy Leave His Famous Comic Strip?

In each of the following, evaluate the variable and use the answers to work out the above riddle.



60 51 95 70 55 98 77 141 97 45 70 98

40 111 81 51 39 63 47 45 63 81 73 111 33 87

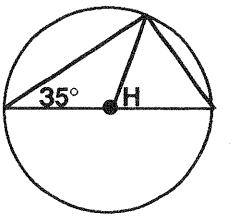
47 63 81 100 77 70 33 55 40 98

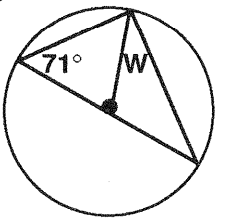
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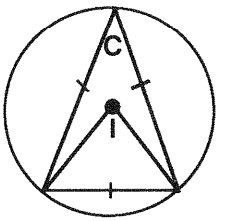


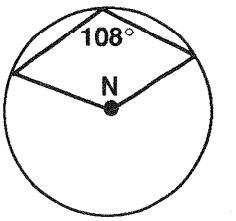
WHAT DO CATS SAY TO EACH OTHER AT CHRISTMAS?

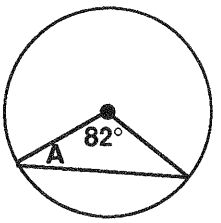
In each of the following circles, find the value of the variable and use the answers in the puzzle below.

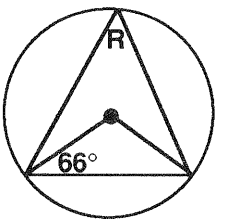


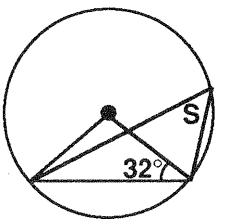


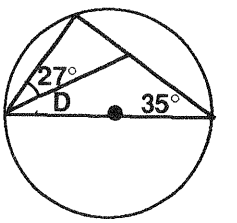


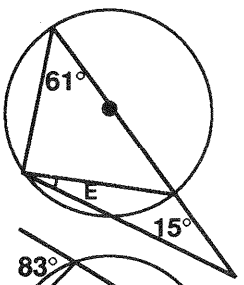


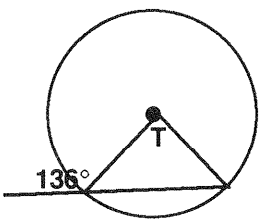


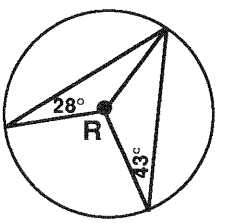


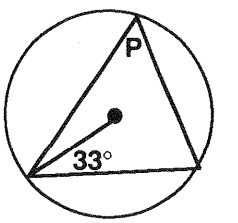


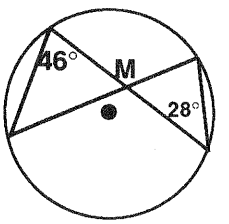


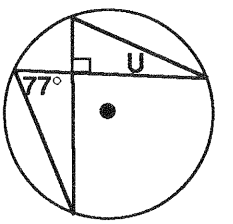


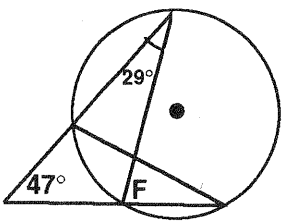












49°	76° 13° 24° 24° 194°	74° 14° 24° 24° 194°
60° 70° 24° 120° 58° 92° 74° 49° 58°		49° 144° 28°
49°	70° 49° 57° 57° 194°	74° 14° 19°
	194° 14° 49° 24°	

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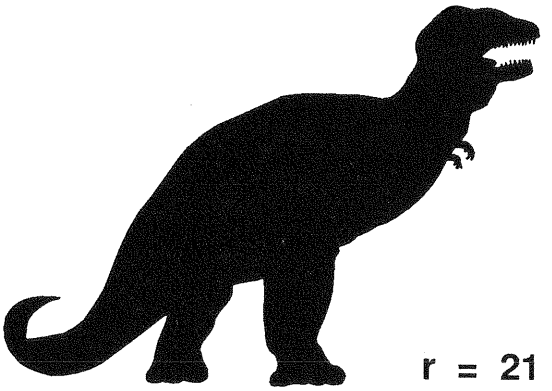
Find the surface area of spheres with the following radii. Join the radius to the correct surface area. The squares the line intersects will help you solve the puzzle.

RADIUS		SURFACE AREA
5 m	✦	✦ 804 m^2
2·8 m	✦	✦ 99 m^2
12 m	✦	✦ 314 m^2
8 m	✦	✦ 154 m^2
7·4 m	✦	✦ 483 m^2
3·5 m	✦	✦ 1810 m^2
8·3 m	✦	✦ 366 m^2
6·2 m	✦	✦ 866 m^2
5·4 m	✦	✦ 1041 m^2
9·1 m	✦	✦ 688 m^2

1	2	3	4	5		6	7	8	9	10	11	12

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WHAT DO YOU GET IF A TYRANNOSAURUS BITES YOU?



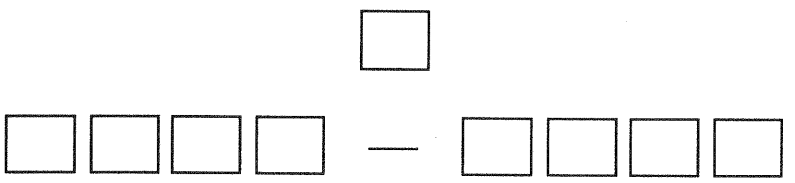
Work out the **VOLUME** of **SPHERES** with the following radii. Round each answer to the nearest unit and cross out the answers in the squares below. The remaining letters will tell you the answer to the puzzle.

RADIUS

r = 21 cm
r = 15 cm
r = 8 cm
r = 2 cm
r = 13 cm
r = 8·9 cm
r = 11·2 cm

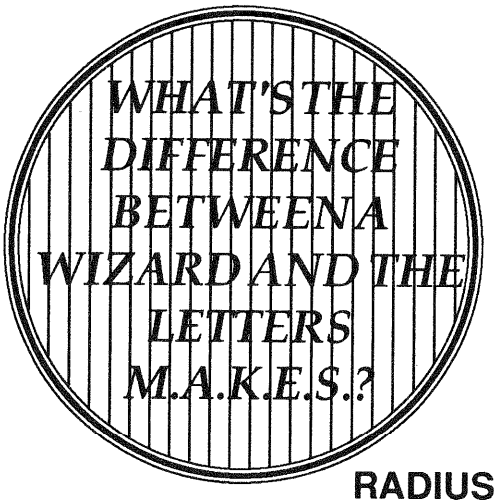
r = 18·3 cm
r = 12·1 cm
r = 19·3 cm
r = 2·7 cm
r = 5·1 cm
r = 3·8 cm

5	5	6	9	3	8	7	9	2	2	8	2	3	4	3	1	4	1	3	7
B	E	C	A	T	N	Z	Y	O	D	D	Y	A	E	I	T	E	O	B	R
2	9	5	3	5	2	5	6	7	1	2	3	0	2	7	4	2	1	9	
P	A	T	S	N	T	O	N	M	O	B	L	S	O	F	O	G	A	S	
3	0	1	1	3	2	1	4	5	6	5	8	8	5	6	4	9	2	0	3
E	T	P	Y	Z	T	S	A	R	O	A	R	G	H	R	E	V	E	G	S





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Find the **VOLUME** and **SURFACE AREA** for spheres with the following radii. Use the answers to interpret the puzzle.

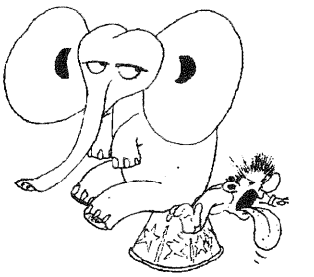
RADIUS	VOLUME	SURFACE AREA
r = 10 cm	E	A
r = 4 cm	S	O
r = 9 cm	M	T
r = 7 cm	E	S
r = 8 cm	L	N
r = 3 cm	N	D
r = 2·8 cm	H	S
r = 8·4 cm	R	K
r = 6·2 cm	E	L
r = 11·8 cm	P	P

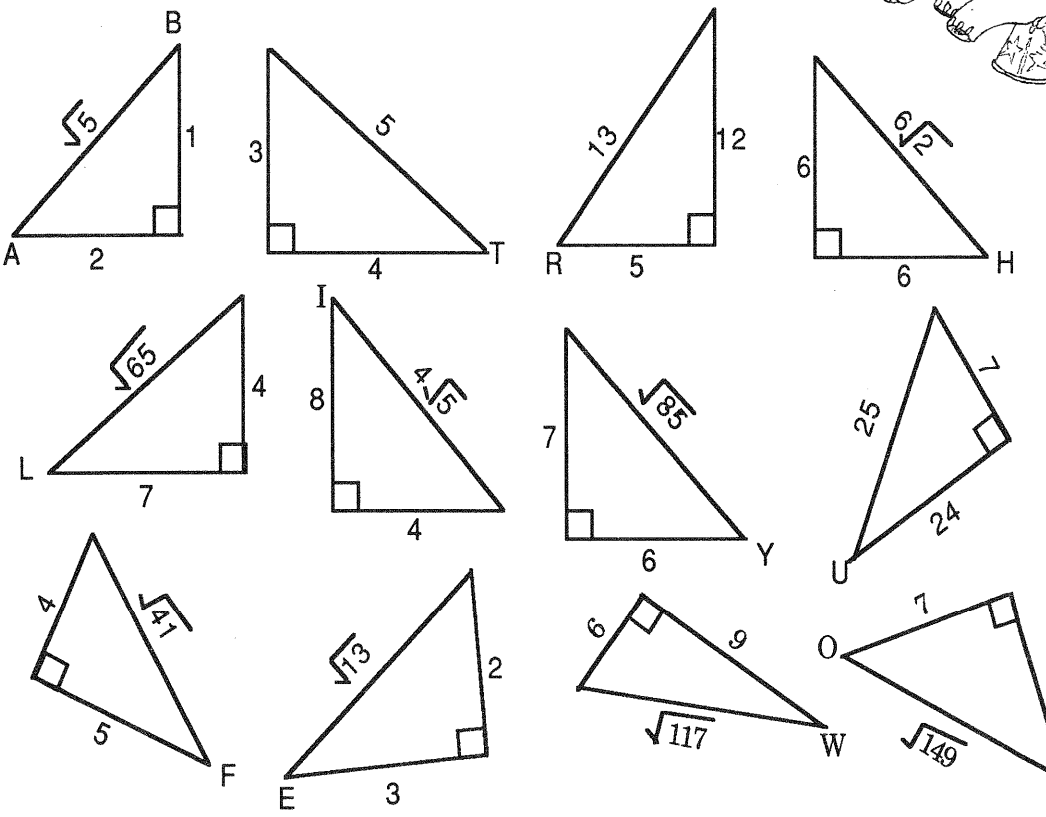
201 cm^2	804 cm^2	4189 cm^3		3054 cm^3	1257 cm^2	887 cm^2	1437 cm^3	268 cm^3		616 cm^2	6882 cm^3	1437 cm^3	2145 cm^3	483 cm^2	268 cm^3		
	1257 cm^2	113 cm^3	113 cm^2		1018 cm^2	92 cm^3	998 cm^3			201 cm^2	1018 cm^2	92 cm^3	998 cm^3	2483 cm^3			
99 cm^2	1750 cm^2	998 cm^3	483 cm^2	2145 cm^3	99 cm^2					3054 cm^3	1257 cm^2	887 cm^2	998 cm^3	268 cm^3			

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WHY IS HAVING NO MONEY LIKE HAVING AN ELEPHANT SITTING ON YOU?

Using the following diagrams of right triangles, find the value for the trigonometric ratios. (Leave your answers as fractions.) Use the variables to decode the puzzle.





sin A sin R cos L sin F tan K

tan B tan H tan Y cos E tan O

cos T tan I cos U cos W



$\frac{3}{\sqrt{13}}$	$\frac{1}{2}$	$\frac{4}{5}$	1	$\frac{3}{\sqrt{13}}$	$\frac{12}{13}$

$\frac{9}{\sqrt{117}}$	$\frac{1}{\sqrt{5}}$	$\frac{7}{6}$

$\frac{7}{6}$	$\frac{10}{7}$	$\frac{24}{25}$	$\frac{12}{13}$	$\frac{3}{\sqrt{13}}$

$\frac{4}{\sqrt{41}}$	$\frac{7}{\sqrt{65}}$	$\frac{1}{\sqrt{5}}$	$\frac{4}{5}$

2	$\frac{12}{13}$	$\frac{10}{7}$	$\frac{7}{10}$	$\frac{3}{\sqrt{13}}$

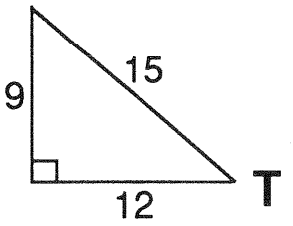
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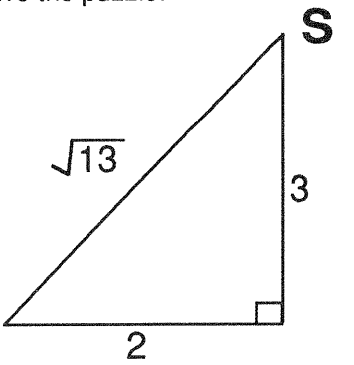


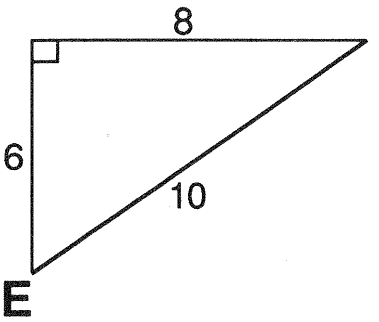
WHAT IS A DOGMA?

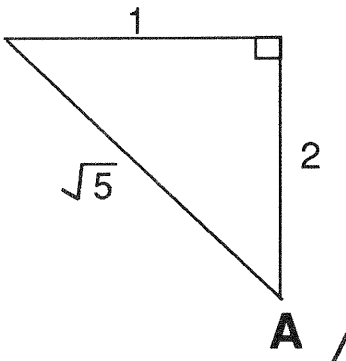
Using the following triangles, find the value of the trigonometric ratios. Convert each to a decimal and give your answer to four decimal places. Use the variables to solve the puzzle.

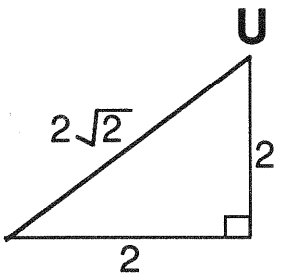
tan	T
cos	S
sin	E
cos	A
sin	U
tan	O
cos	I
sin	P
tan	H
sin	I
cos	R
tan	M
cos	Y

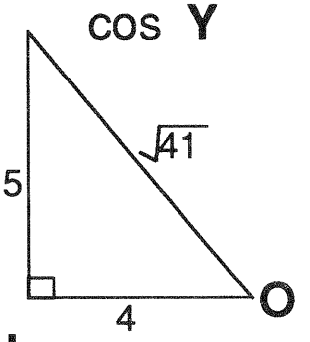


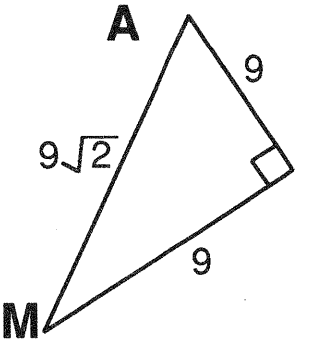


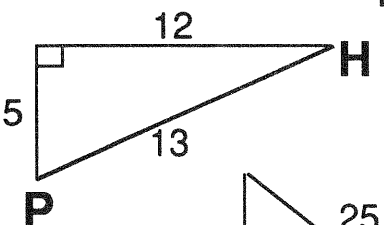


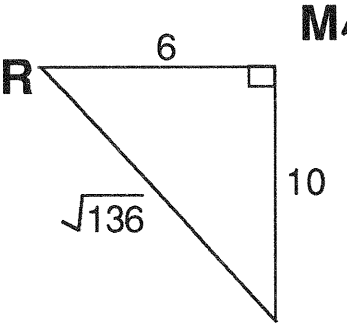


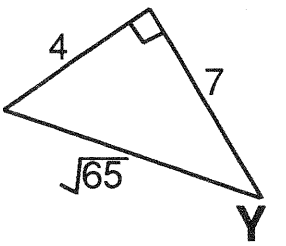


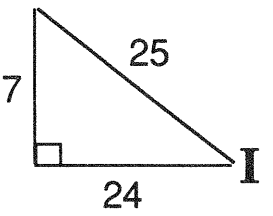








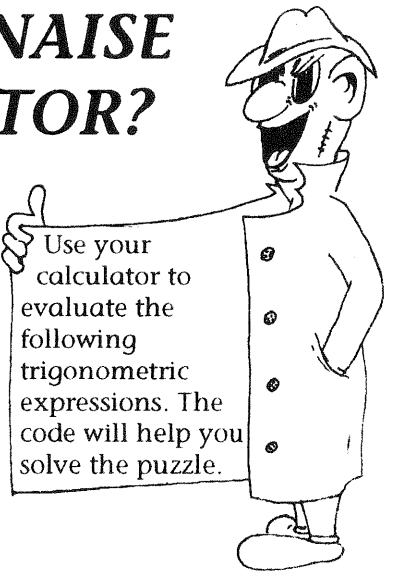




0·2800	0·7500		0·9600	0·8321		0·8944		0·9231	0·7071	0·9231	0·9231	0·8682		0·8321		1·0000	1·2500	0·7500	0·4167	0·8000	0·5145
													'								

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WHAT DID THE MAYONNAISE SAY TO THE REFRIGERATOR?



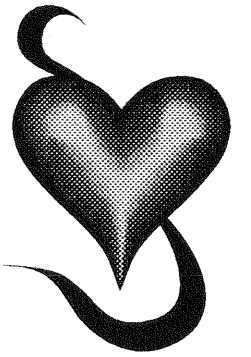
1. sin 37°
2. cos 28°
3. tan 42°
4. tan 19·28°
5. cos 32·17°
6. cos 87°
7. sin 49·78°
8. tan 67°
9. tan 5·63°
10. sin 72°
11. tan 68·17°
12. sin 27·09°
13. sin 77·27°
14. cos 29·73°
15. sin 58°54'
16. tan 68°30'
17. tan 15°21'
18. cos 70°48'
19. sin 53°36'
20. tan 78°18'
21. cos 20°42'
22. cos 81°32'

CODE

0·0523	– T	0·8563	– D
0·9004	– O	0·8049	– S
0·9754	– I	0·9511	– O
0·7636	– H	0·1472	– G
0·8684	– M	2·5386	– R
2·3559	– E	4·8288	– I
0·6018	– C	0·3498	– S
0·4554	– R	0·3289	– S
0·0986	– D	2·4964	– O
0·2745	– E	0·8465	– E
0·8829	– L	0·9354	– N

1	2	3	4	5		6	7	8		9	10	11	12	—							

WHAT DO SQUIRRELS GIVE EACH OTHER ON VALENTINE'S DAY?



For each of the following trigonometric equations, find the size of the angle.

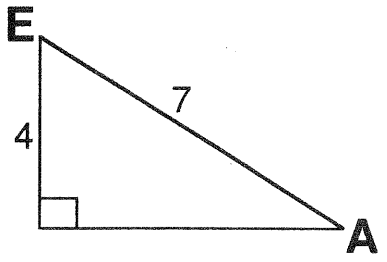
sin x = 0·4848	U	57°
sin x = 0·7548	M 7	49°
cos x = 0·6157	10 N	22°
cos x = 0·5446	5 G 9	47°
tan x = 0·4040	4 E 8 F	29°
sin x = 0·9962	6 T	85°
tan x = 1·9626	2 1	52°
cos x = 0·9563	O	78°
cos x = 0·6820	3 T	63°
sin x = 0·5878	S 12	36°
tan x = 4·7046	11 R	17°

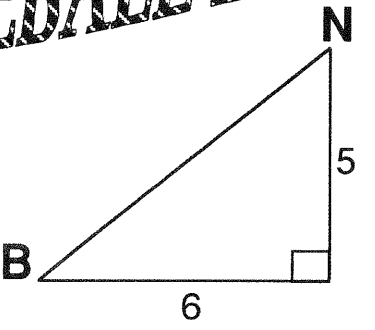
1	2	3	4	5	6	–	7	8	–	9	10	11	12

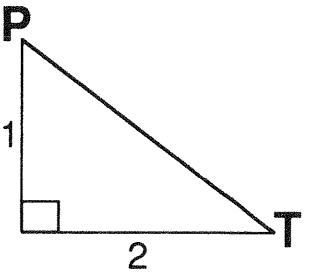
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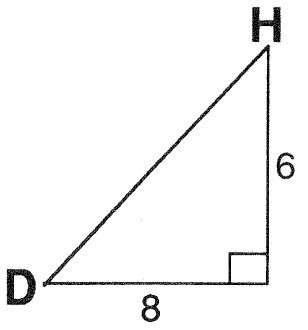
HOW IS BASEBALL LIKE PANCAKES?

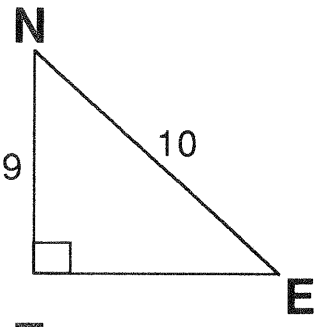
Use the dimensions in each of the following right triangles to determine the trigonometric ratio and then to find the size of the angle.

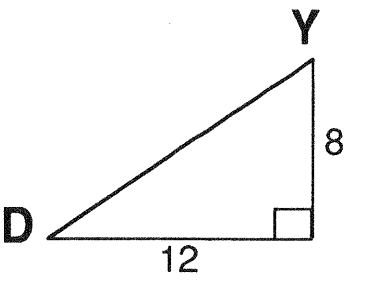


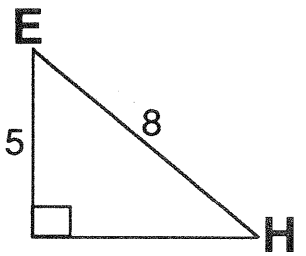


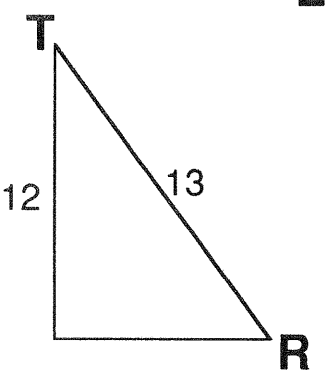


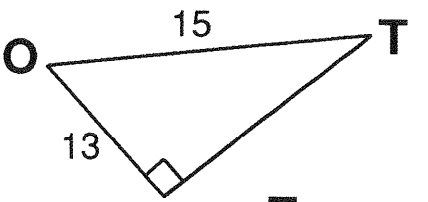


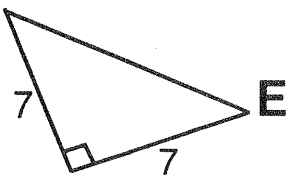


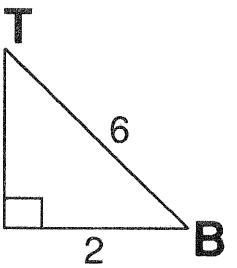


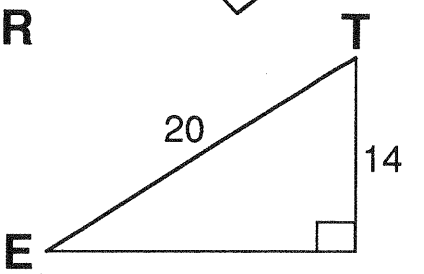






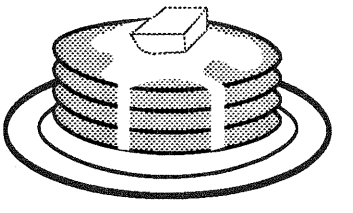






	60·07°	38·68°	45·00°	56·31°	39·81°	29·93°	19·47°	53·13°	36·87°	51·32°	63·43°	64·16°	25·84°	33·69°

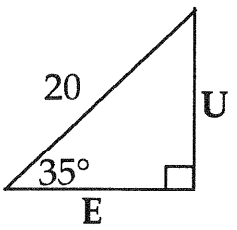
29·93°	50·19°	26·57°	38·68°	55·15°	70·53°	34·85°	45·57°	22·62°	44·43°	67·38°

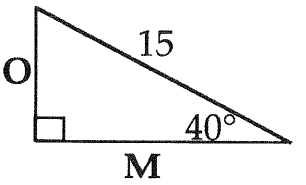


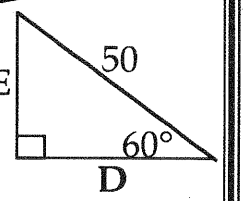
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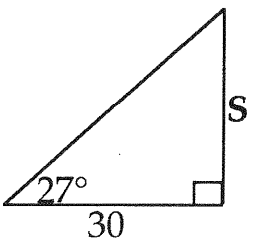
WHEN WAS SPAGHETTI INVENTED?

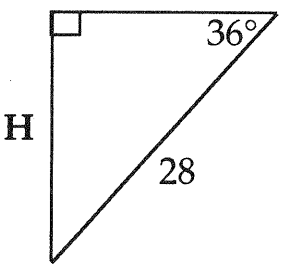
Find the height of the side indicated by the variable in each of the following right triangles. Use the code to solve the puzzle.

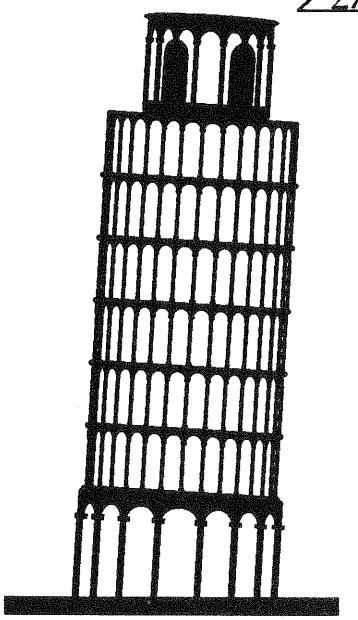


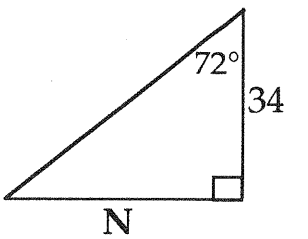


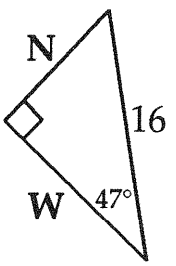


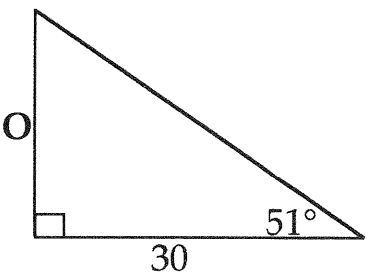


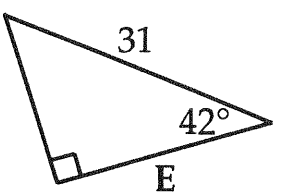


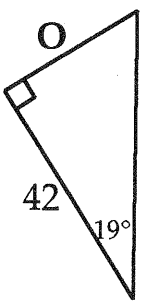


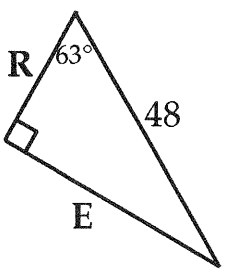


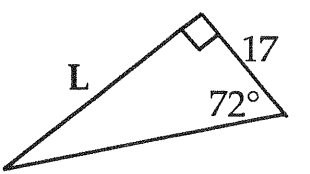












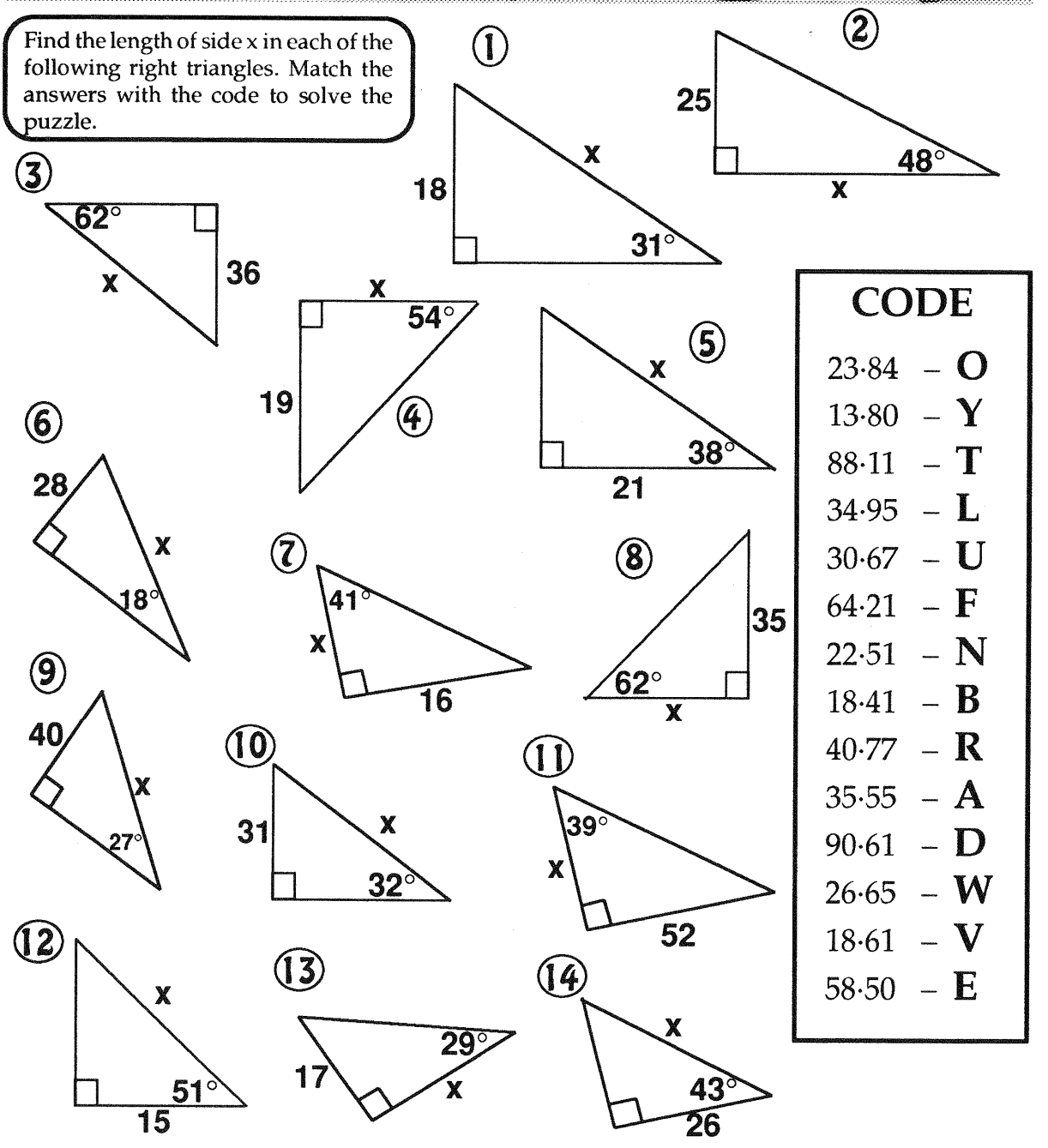
10·91	16·46	23·04	11·70	15·29	9·64	11·49	42·77	37·05	104·64	23·04	11·47	15·29	43·30	25

16·46	43·30	21·79	104·64	14·46	37·05	25	52·32	16·38

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WHAT DID THE CABBAGE SAY TO HER SON WHEN HE TOLD A LIE?

Find the length of side x in each of the following right triangles. Match the answers with the code to solve the puzzle.



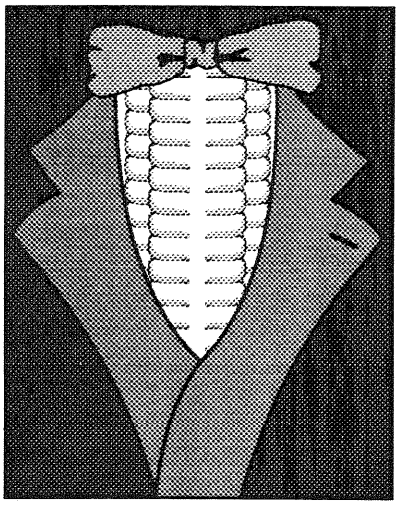
CODE

23·84	–	O
13·80	–	Y
88·11	–	T
34·95	–	L
30·67	–	U
64·21	–	F
22·51	–	N
18·41	–	B
40·77	–	R
35·55	–	A
90·61	–	D
26·65	–	W
18·61	–	V
58·50	–	E

4 12 13 6 ' 7 10 9 9 10 3 ' 9 13 3 2

12 8 10 3 ' 14 ' 2 10 5 ' 1 10 14 11

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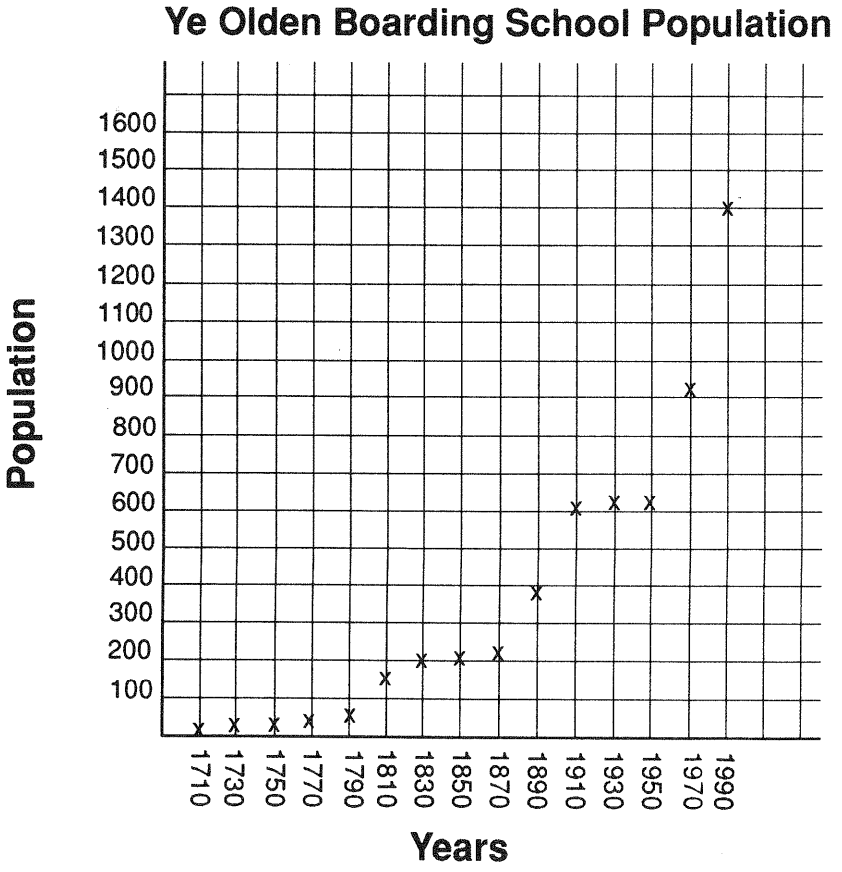
WHAT DID THE MOTHER CORN TELL HER BABY ABOUT WHERE IT CAME FROM?

CODE

100 – A	490 – E	300 – Y
1180 – T	50 – U	1280 – K
750 – R	620 – G	175 – H
590 – B	200 – L	20 – O
220 – S		

Estimate the school population for the following years:

1. 1800
2. 1980
3. 1880
4. 1820
5. 1780
6. 1900
7. 1720
8. 1940
9. 1960
10. 1985
11. 1908
12. 1828
13. 1860



2	
4	
6	
2	
1	
10	
11	
9	
7	
5	
8	
4	
2	
3	
7	
5	

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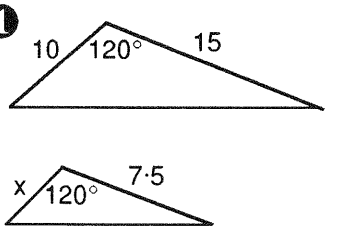
WHAT KIND OF VEGETABLE DO THEY USE TO MAKE LASSIE'S DOG BISCUITS?

For each pair of similar triangles, find the value of the variables.

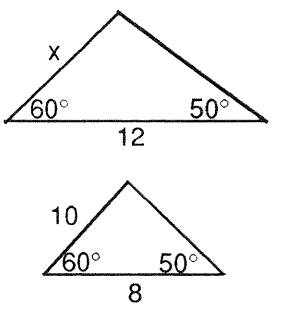
CODE

5	– U
3·3̇	– F
14	– I
7·5	– L
15	– R
10	– E
25	– C
20	– O

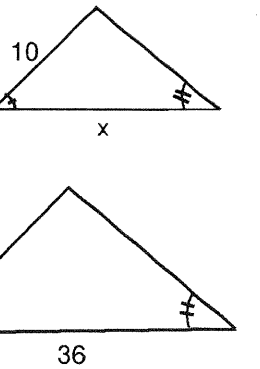
❶



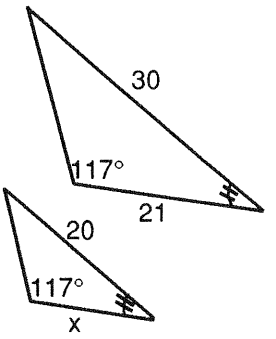
❷



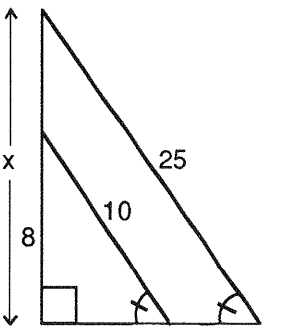
❸



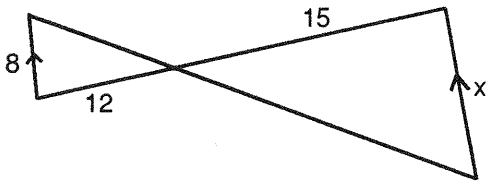
❹



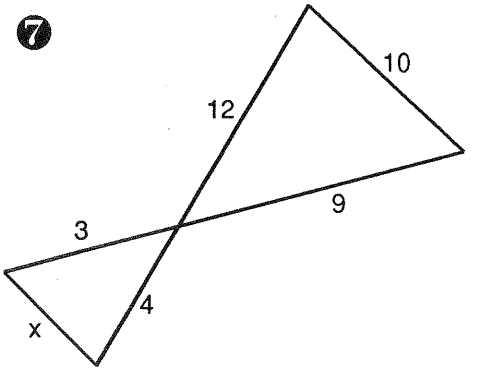
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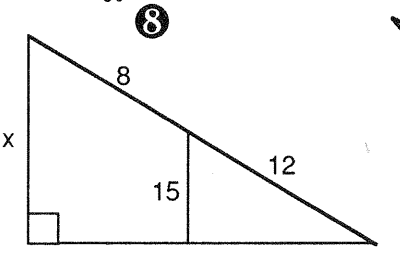
❻



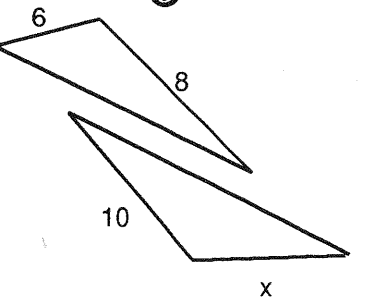
❼

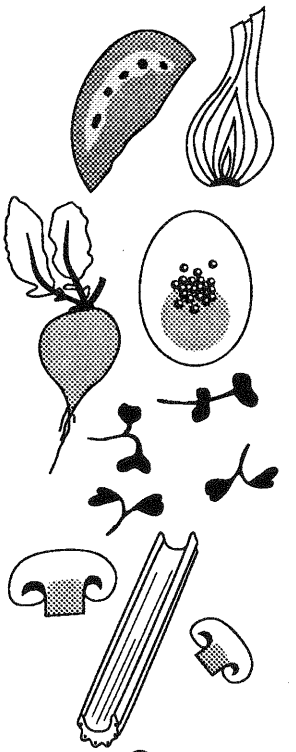


❽



❾





8	5	9	9	4	6	–	7	9	5	1	2

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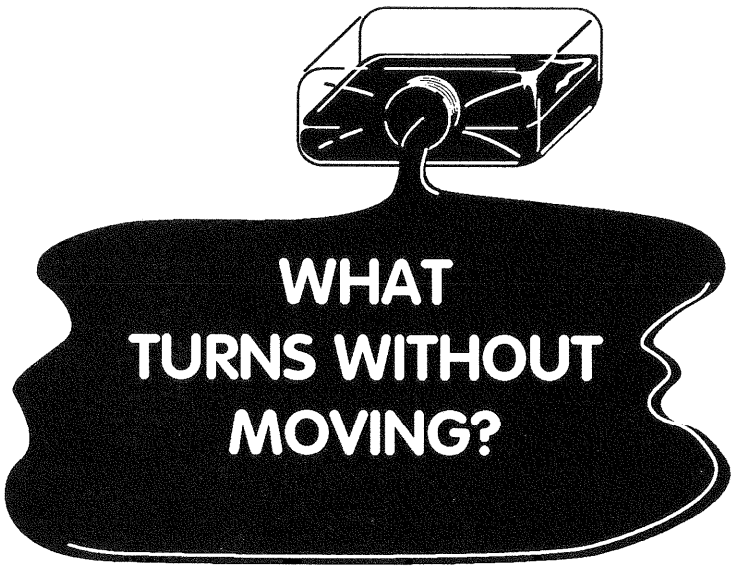
WHAT DID ONE MELON SAY TO THE OTHER MELON ON VALENTINES DAY?

Convert each of the following rates to km/hr.

10 m/sec	– V
30 m/min	– W
1·5 km/min	– Y
200 cm/sec	– U
150 m/min	– O
175 m/2 min	– E
25 m/2 sec	– M
200 m/3 min	– E
4000 m/20 min	– O
160 m/15 sec	– Y
210 m/6 sec	– E
360 m/90 sec	– O
2·1 km/2·5 min	– L
61 m/1·5 sec	– E
18 m/2·4 sec	– N
12·8 km/15 min	– D
300 m/19·2 sec	– H

56·25	14·4	27	126	38·4	51·2	146·4	1·8	90	9	7·2	50·4	12	36	4	45	5·25

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WHAT TURNS WITHOUT MOVING?

In a road survey, the number of various types of vehicles were recorded in the following table. If this information was to be represented on a circle graph, how many degrees would each category need?

	VEHICLE	NUMBER
1.	GMH	190
2.	FORD	210
3.	TOYOTA	250
4.	NISSAN	80
5.	PUGEOT	9
6.	MITSUBISHI	45
7.	CHRYSLER	48
8.	FIAT	36
9.	MERCEDES	5
10.	ROLLS ROYCE	2
11.	FERRARI	0
12.	RENAULT	11
13.	BMW	14

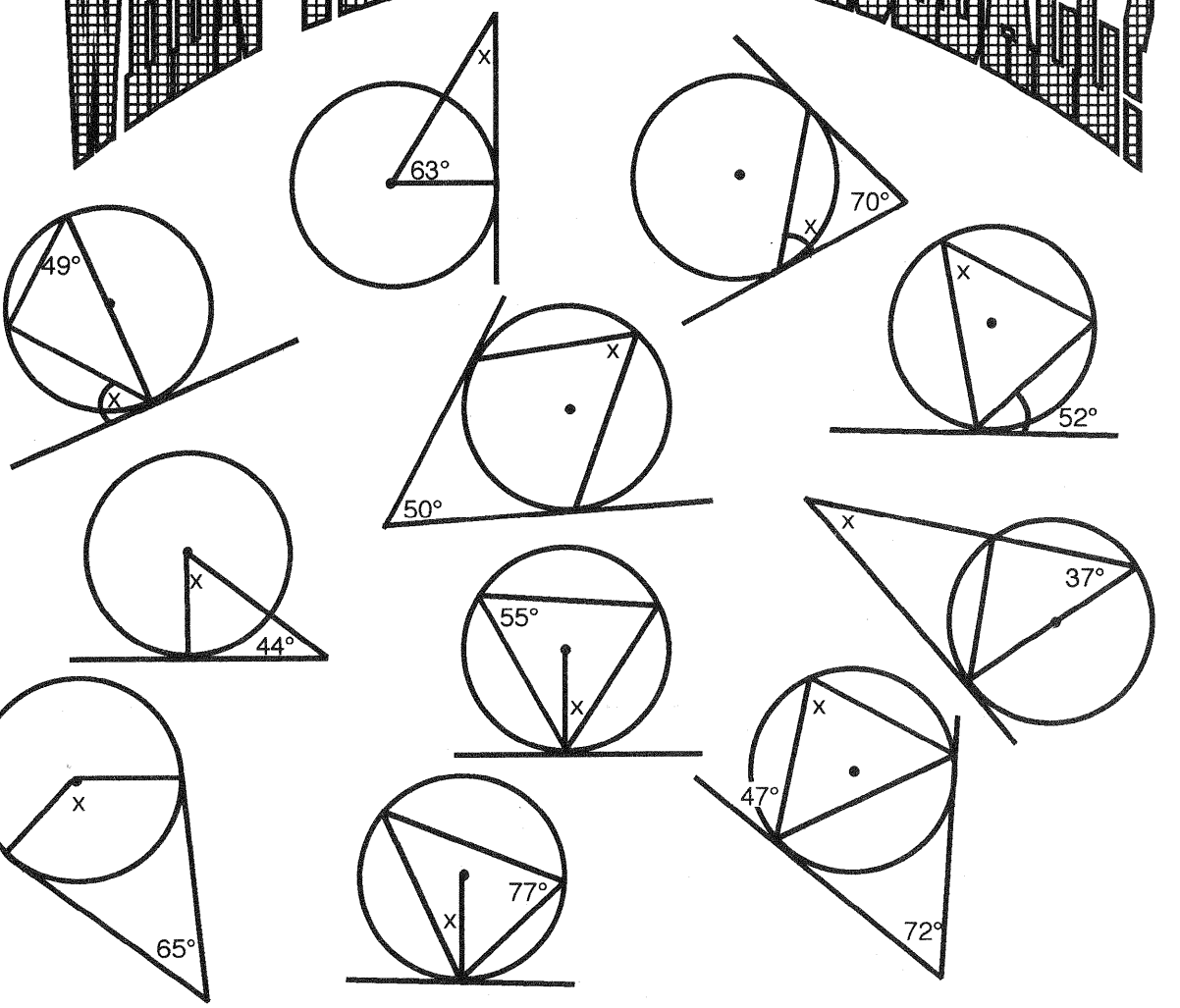
CODE	
32°	– I
19·2°	– N
84°	– T
4·4°	– K
5·6°	– M
18°	– T
76°	– U
2°	– L
14·4°	– U
100°	– R
3·6°	– S
0°	– O

				—											
13	*4*	*9*	*12*		*4*	*6*	*2*	*1*	*3*	*7*	*5*	*5*	*11*	*8*	*3*

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Find the size of the angle indicated by the variable in each of the following circles. Find the answers in the boxes below and cross out the adjacent letters. The remaining letters in order will give the answer to the riddle.

WHAT DO DAIRY WORKERS YELL WHEN THEY PLAY BASEBALL?



1	1	5	9	0	5	3	2	2	7	2	3	5	8	5	2
A	C	O	B	U	M	R	T	S	O	T	Y	A	E	B	T
4	9	8	6	5	4	6	5	4	1	3	9	5	5	6	
D	E	R	A	M	X	Q	U	O	V	D	U	R	P	P	

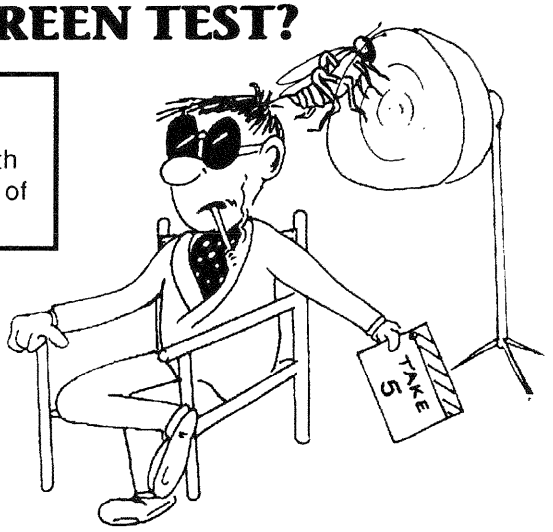
_____ _____ _____ _____ _____ _____ — _____ _____

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WHAT DID THE PRODUCER TELL THE BUGS AFTER THE SCREEN TEST?

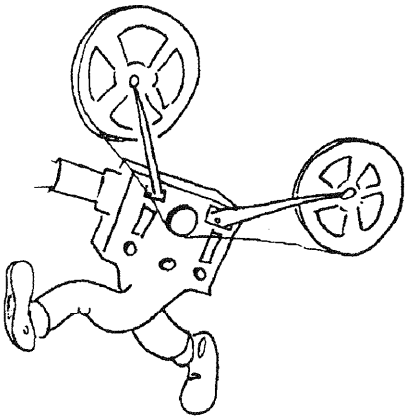


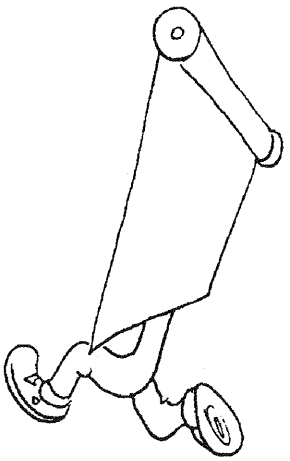
In each of the following problems you are required to find the angle subtended at the centre of the earth by the two points on the same line of longitude or latitude.



W	38°N, 42°E	and	67°N, 42°E
U	16°S, 75°W	and	84°S, 75°W
O	34°S, 28°W	and	45°N, 28°W
C	27°N, 126°W	and	27°N, 15°W
L	19°S, 108°E	and	19°S, 47°W
S	42°S, 63°E	and	75°N, 63°E

E	51°N, 32°W	and	51°N, 177°W
N	0°N, 75°W	and	0°N, 92°E
Y	18°N, 125°W	and	58°S, 125°W
A	26°S, 54°E	and	26°S, 17°W
T	77°N, O°E	and	64°S, O°E
W	19°S, 114°W	and	19°S, 180°W





L	14°30'N, 76°W	and	28°30'S, 76°W
D	49°S, 13°30'W	and	49°S, 76°30'E
C	27°15'S, 33°E	and	27°45'N, 33°E
A	42°21'N, 15°W	and	17°39'S, 15°W
R	64°S, 8°29'W	and	64°S, 162°29'W

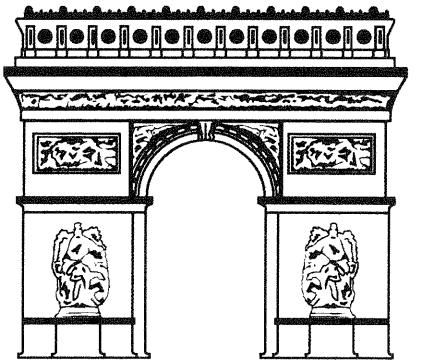
90	79	167	141		111	154	60	29	43		68	117,
												,
	'											
66	145	43	155	55	154	71	29	155		76	79	68





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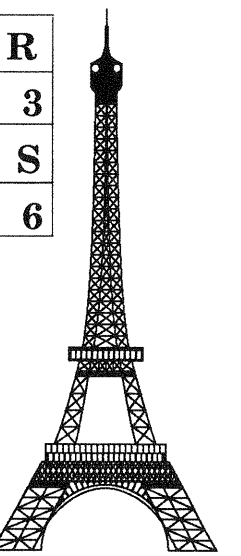
WHAT ARE FRENCH TICKS CALLED?



Find the distance between the following pairs of points on the earth's surface. Use 6335 km as the radius of the earth. Give your answers to the nearest km. Find the answers in the boxes below; cross out the adjacent letter. The remaining letters in order will answer the riddle.

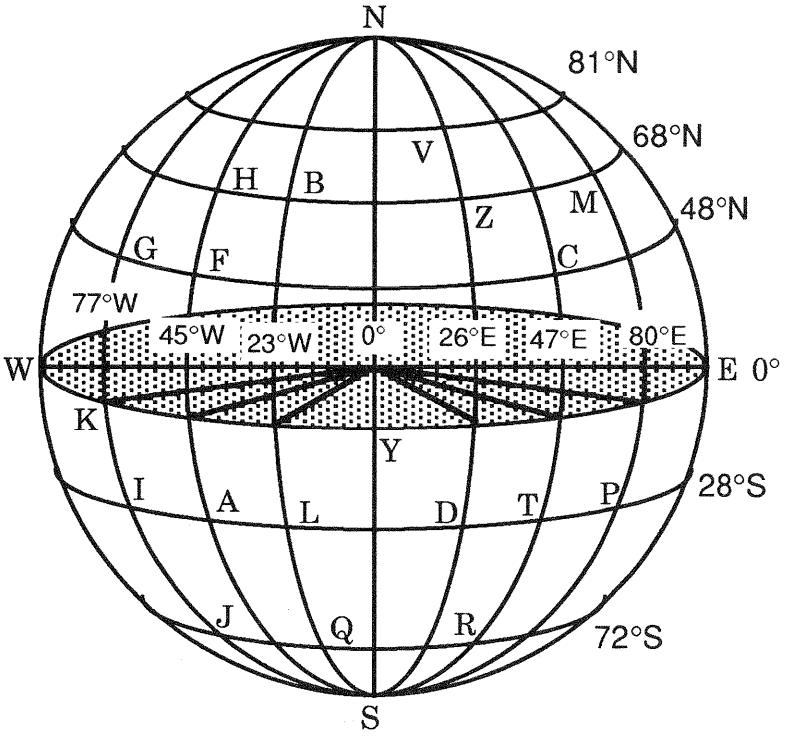
33°N, 45°E	and	13°N, 45°E
28°N, 17°W	and	56°N, 17°W
72°N, 115°E	and	16°S, 115°E
25°S, 39°W	and	21°N, 39°W
48°S, 134°W	and	62°N, 134°W
64°S, 0°E	and	73°N, 0°E
0°S, 34°W	and	0°S, 18°E
15°S, 45°E	and	76°N, 45°E
2°S, 173°W	and	24°N, 173°W
0°N, 43°W	and	0°N, 62°E

A	M	O	R	I	N	G	A	S	P	Q	U	I	X	A	D	O	N	A	R
2	2	1	1	1	1	6	0	9	0	5	0	8	6	4	9	7	3	0	3
P	A	Y	S	I	S	T	P	D	E	F	G	E	R	O	M	T	O	U	S
5	7	4	9	0	2	1	2	1	6	2	1	5	1	4	8	3	0	9	6
S	T	E	A	S	-	I	T	B	C	A	R	E	S						
1	0	0	6	2	2	3	2	2	8	7	5	4	7						



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WHY DO LICE FAMILIES GET ALONG SO WELL?



Find the size of the angle subtended at the centre of the earth by each of the following pairs of points.

1. LD
2. IP
3. FA
4. JQ
5. TP
6. HB
7. VZ
8. CR
9. LQ
10. BM
11. AP
12. KY
13. DV
14. JR
15. MP

CODE

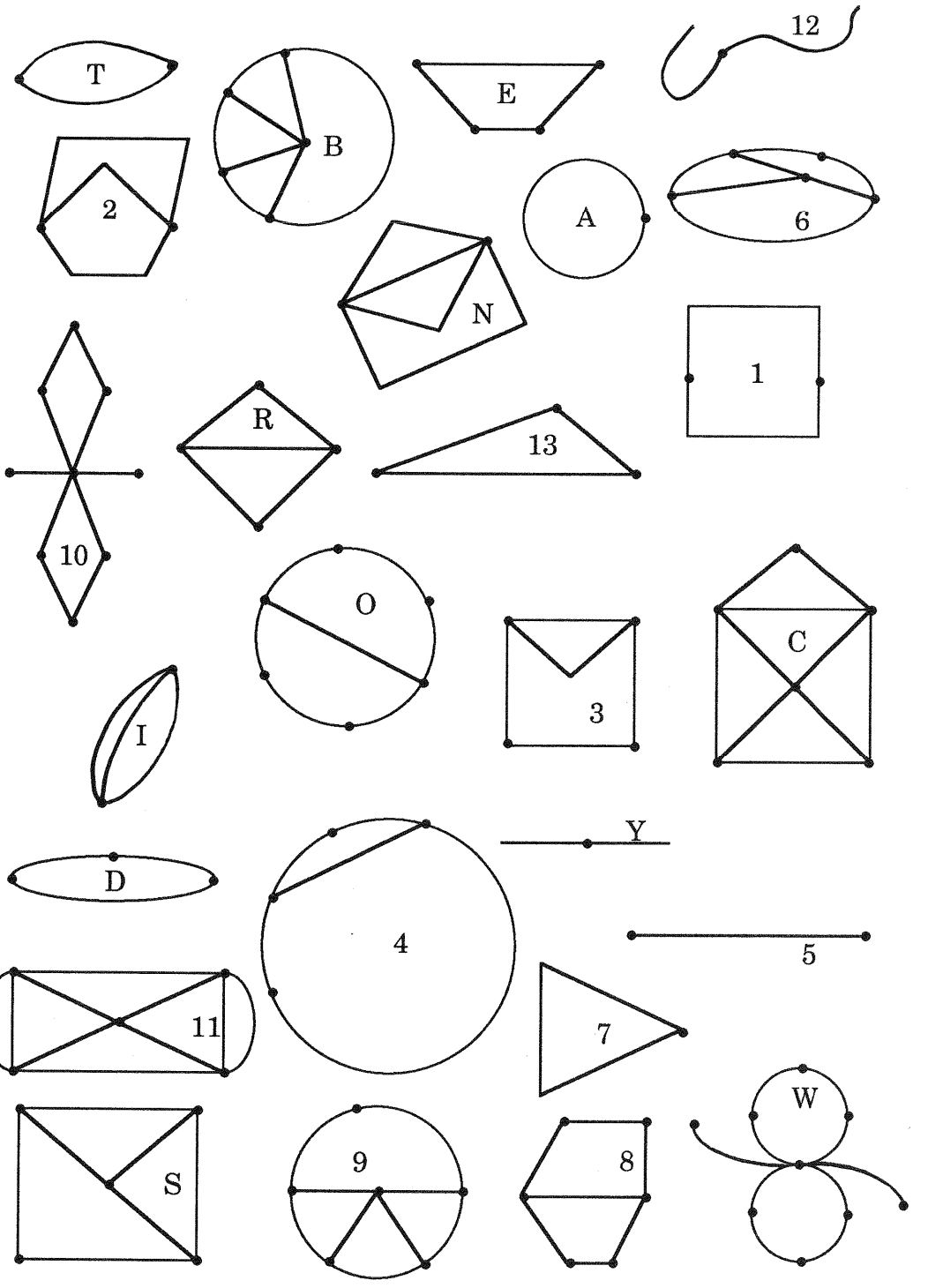
54°	– **Y**	157°	– **E**
103°	– **T**	13°	– **H**
49°	– **E**	125°	– **T**
77°	– **S**	76°	– **A**
22°	– **L**	109°	– **E**
44°	– **R**	33°	– **I**
96°	– **N**	124°	– **C**
		120°	– **O**

10	7	13	4	3	9	1	14	6	8	12	2	–	15	5	11

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WHERE DO SPIDERS GO TO LEARN NEW WORDS?

For each of the following, find two shapes that are topologically equivalent.



			–															
10	3	11		6	1	3	4	6	13	2	9	1	2	8	5	7	4	12

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MATH O MAGIC 5·3 SOLUTIONS

1. THEY ARE THE LAST ONES TO LET YOU DOWN
2. WITH EXTREME CAUTION
3. WELL NO NOOSE IS GOOD NOOSE
4. YOU ROCKET
5. WITH THE BEST VICIOUS OF THE SEASON
6. PLEASE GO AWAY
7. BECAUSE HE WAS TIRED OF WORKING FOR PEANUTS
8. A FURRY MERRY CHRISTMAS AND A HAPPY NEW YEAR
9. NAVAL ORANGES
10. A DINO – SORE
11. ONE MAKES SPELLS AND THE OTHER SPELLS MAKES
12. EITHER WAY YOU'RE FLAT BROKE
13. IT IS A PUPPY'S MOTHER
14. CLOSE THE DOOR – I'M DRESSING
15. FORGET – ME – NUTS
16. THEY BOTH RELY ON THE BATTER
17. WHEN SOMEONE USED HER NOODLE
18. YOU'D BETTER TURN OVER A NEW LEAF
19. THE STALK BROUGHT YOU
20. COLLIE – FLOUR
21 HONEYDEW YOU LOVE ME
22. MILK – IT TURNS SOUR
23. BUTTER UP
24. DON'T CRAWL US, WE'LL CRAWL YOU
25. THEY ARE CLOSE – NIT
26. PARIS – ITES
27. WEB–STER'S DICTIONARY

